Lesson 7 – Octal to Decimal Conversion

NUMBER SYSTEM CONVERSION

Octal to Decimal Conversion

A step-by-step mathematical breakdown for converting base-8 numbers to base-10 representations.

Base-8 $\rightarrow$ Base-10

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General Formula: Decimal Value = (dn × 8n) + ... + (d1 × 81) + (d0 × 80)

CORE CONCEPTS

Understanding Positional Notation

Octal System (Base-8)

Uses eight distinct digits: 0 through 7. Each positional rank represents a power of 8.

Decimal System (Base-10)

Standard human counting system using digits 0 through 9 based on powers of 10.

Conversion Rule

Multiply each octal digit by 8n, where n is the digit's position index from right to left.

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Powers of 8 Reference: 80 = 1 | 81 = 8 | 82 = 64 | 83 = 512 | 84 = 4096

METHODOLOGY

4-Step Conversion Process

STEP 01

Identify Digits

Isolate every individual digit of the octal number.

STEP 02

Assign Positions

Assign positional powers of 8 starting from 0 on the right.

STEP 03

Multiply Products

Multiply each digit by its corresponding power of 8.

STEP 04

Sum Results

Add all products together to obtain the decimal value.

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PRACTICAL APPLICATION

Worked Example: Convert (157)8 to Decimal

Breakdown Table

Summing up the components:

64 + 40 + 7 = 111

RESULT

(157)8 = (111)10

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Conversion Complete

Any octal number can be reliably converted to decimal by evaluating its positional base-8 powers.